

Curriculum Vitae

Harvansh Dandelia

Personal Information

Name: Harvansh Dandelia
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Education

Degree	Institution	GPA	Year
Doctor of Philosophy (PhD) in Mechanical Engineering (Thesis defended successfully)	Indian Institute of Technology Gandhinagar	9.11/10	2020-2025
Master of Technology (Mechanical Engineering)	Indian Institute of Technology Gandhinagar	7.79/10	2018-2020
Bachelor of Technology (Mechanical Engineering)	National Institute of Technology Raipur	8.34/10	2014-2018

Doctoral (PhD) Experience

Linear Dynamics of Pulsating Taylor–Couette flow: Instabilities and Transition. 2020-2025

- Developed a Floquet-based modal stability framework for time-periodic Taylor–Couette flow with pulsating inner cylinder.
- Computed neutral stability curves and identified subharmonic–harmonic transition mechanisms across parameter regimes.
- Performed non-modal transient growth analysis using direct–adjoint optimization to quantify optimal perturbation amplification.
- Investigated the influence of radius ratio, steady and oscillatory Reynolds numbers, pulsation frequency (Womersley number), and axial flow on stability characteristics.
- Implemented high-order Chebyshev spectral discretization and eigenvalue solvers for large-scale stability problems.

Research Experience

Research Associate-II

Oct 2025 – Present

Indian Institute of Technology Gandhinagar

- Continuing research on hydrodynamic stability and transient growth.
- Focusing on axisymmetric boundary-layer stability.

Laboratoire de Mécanique des Fluides et d'Acoustique (LMFA), Lyon, France

- Collaborated with Prof. Benoît Pier and Prof. Frédéric Alizard on the linear stability of modulated rotating shear flows.
- Performed Floquet stability analysis using Fortran-based solvers and Python automation.
- Manuscript under preparation for submission to a peer-reviewed journal.

Master's Research

M.Tech Thesis: *Control of Growth of Instabilities in Taylor–Couette Flow*

2018–2020

- Developed an LQR-based optimal flow control framework using state-space formulation.
- Verified analytical derivations using Mathematica; implemented numerical simulations in MATLAB.
- Demonstrated significant reduction in optimal transient energy growth.
- Published in *Physics of Fluids*.

Publications

Published

- **Dandelia, H.**, Kant, R., & Narayanan, V. (2024). Control of linear instabilities in two-dimensional airfoil boundary layers. *Fluid Dynamics Research*. DOI
- **Dandelia, H.**, Kant, R., & Narayanan, V. (2022). Optimal control of growth of instabilities in Taylor–Couette flow. *Physics of Fluids*, 34(4). DOI

Manuscripts in Preparation

- **Dandelia, H.**, Pier, B., Alizard, F., Narayanan, V. Subharmonic–harmonic transition in modulated Taylor–Couette flow.
- **Dandelia, H.**, Narayanan, V. Optimal growth of instabilities in pulsating Taylor–Couette flow.

Conference Presentations

- *1st European Fluid Dynamics Conference (EFDC 2024)*, Aachen, Germany (Sept 16–20, 2024). Study titled “Control of instabilities in pulsating channel flow”.
- *26th International Congress of Theoretical and Applied Mechanics (ICTAM 2024)*, Daegu, South Korea (2024). Presented “Optimal growth of instabilities in pulsating Taylor–Couette flow”.
- *European Fluid Mechanics Conference (EFMC 2022)*, Athens, Greece (Sept 13–16, 2022). Presented “Control of growth of instabilities in axisymmetric boundary layer”.
- *Applied Mathematics Symposium: Instability and Flow Transition (IITG–IISc)*, India (Nov 11, 2022). Presented “Control of growth of perturbation in airfoil boundary layers”.
- *APS Division of Fluid Dynamics Meeting*, USA (2021). Narayanan, V., Kant, R., & Dandelia, H. “Control of optimal transient growth on high AR trapezoidal wing boundary layers”.
- *Mathematics and Physics of Fluids*, IIT Gandhinagar, India (Nov 1–3, 2021). Presented “Optimal control of growth of instabilities in Taylor–Couette flow”.

Grants and Awards

- IIT Gandhinagar Overseas Research Exposure Fellowship (Research at LMFA, Lyon, France).
- ITS-ANRF (International Travel support, Anusandhan National Research Foundation) Travel Grant for EFMC 2022, Athens, Greece.
- Publication Award (USD 250) for M.Tech research paper, IIT Gandhinagar.
- Professional Certificate in Scientific Writing, IIT Gandhinagar.

Technical Skills

- **Numerical Methods:** Chebyshev spectral methods, Floquet theory, Arnoldi eigenvalue solvers, direct-adjoint optimization, transient growth analysis, LQR control.
- **Programming:** Fortran (HPC simulations), Python (automation and data analysis), MATLAB.
- **Tools:** Linux HPC environments, SLURM.
- **Languages:** English (fluent), Hindi (native), Spanish (basic).

Academic Service

- Reviewer for *Physics of Fluids* (5 manuscripts reviewed).
- ORCID: 0000-0002-9576-4286

References

- Prof. Vinod Narayanan** Professor, Department of Mechanical Engineering
Indian Institute of Technology Gandhinagar, India
Email: vinod@iitgn.ac.in
- Prof. Benoît Pier** Directeur de Recherche, CNRS
Laboratoire de Mécanique des Fluides et d'Acoustique (LMFA)
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- Prof. Frédéric Alizard** Professor
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